

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Project Title	Script-Controlled ACL – Restrict Record Access Based on Field Value
Team ID	SWTID-2026-1695
Project Name	Skill Wallet

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User & Role Management (Milestone-1)	Create test user records for different departments Create custom roles: u_swt_admin, u_swt_user
FR-2	Table Creation (Milestone-2)	Create custom table “Skill Wallet Request” (u_skill_wallet_request) Define fields: Employee, Department, Skill Name, Wallet Points, Status
FR-3	Access Control – Read (Milestone-3)	Configure Script-Controlled ACL on the Read operation, checking Department field value against the logged-in user
FR-4	Access Control – Create (Milestone-4)	Configure Script-Controlled ACL on the Create operation, checking Department field value against the logged-in user
FR-5	Access Control – Write (Milestone-5)	Configure Script-Controlled ACL on the Write/Update operation, checking Department field value against the logged-in user
FR-6	Access Control – Delete (Milestone-6)	Configure Script-Controlled ACL on the Delete operation, restricted to the u_swt_admin role only

Non-Functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Uses the standard ServiceNow list/form view; no additional custom UI is required for end users to interact with Skill Wallet records.
NFR-2	Security	Central requirement of the project — dynamic, field-value-based Script-Controlled ACLs prevent any department's user from viewing, creating, updating, or deleting another department's records.
NFR-3	Reliability	ACL scripts are evaluated by the ServiceNow platform on every access attempt (UI, list, and API), ensuring the restriction is enforced consistently every time.
NFR-4	Performance	Each ACL script performs a single lightweight GlideRecord lookup on sys_user, adding negligible overhead to record access.
NFR-5	Availability	Implemented as a native ServiceNow platform feature (ACL engine); available whenever the ServiceNow instance is available, with no external dependency.
NFR-6	Scalability	The same script-controlled ACL pattern can be reused on additional tables or fields with minimal modification, without redesigning the security model.